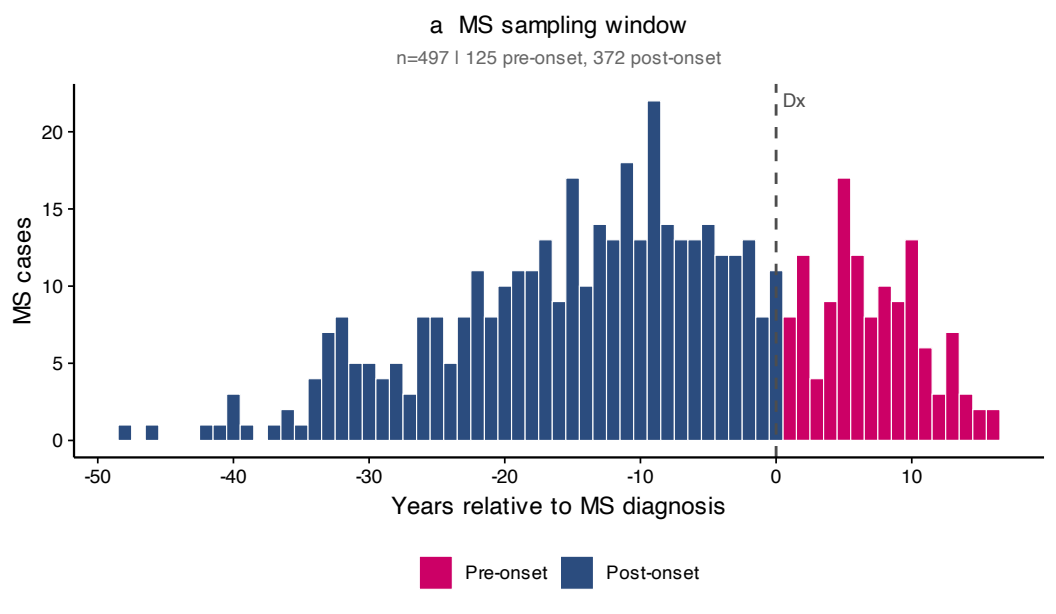
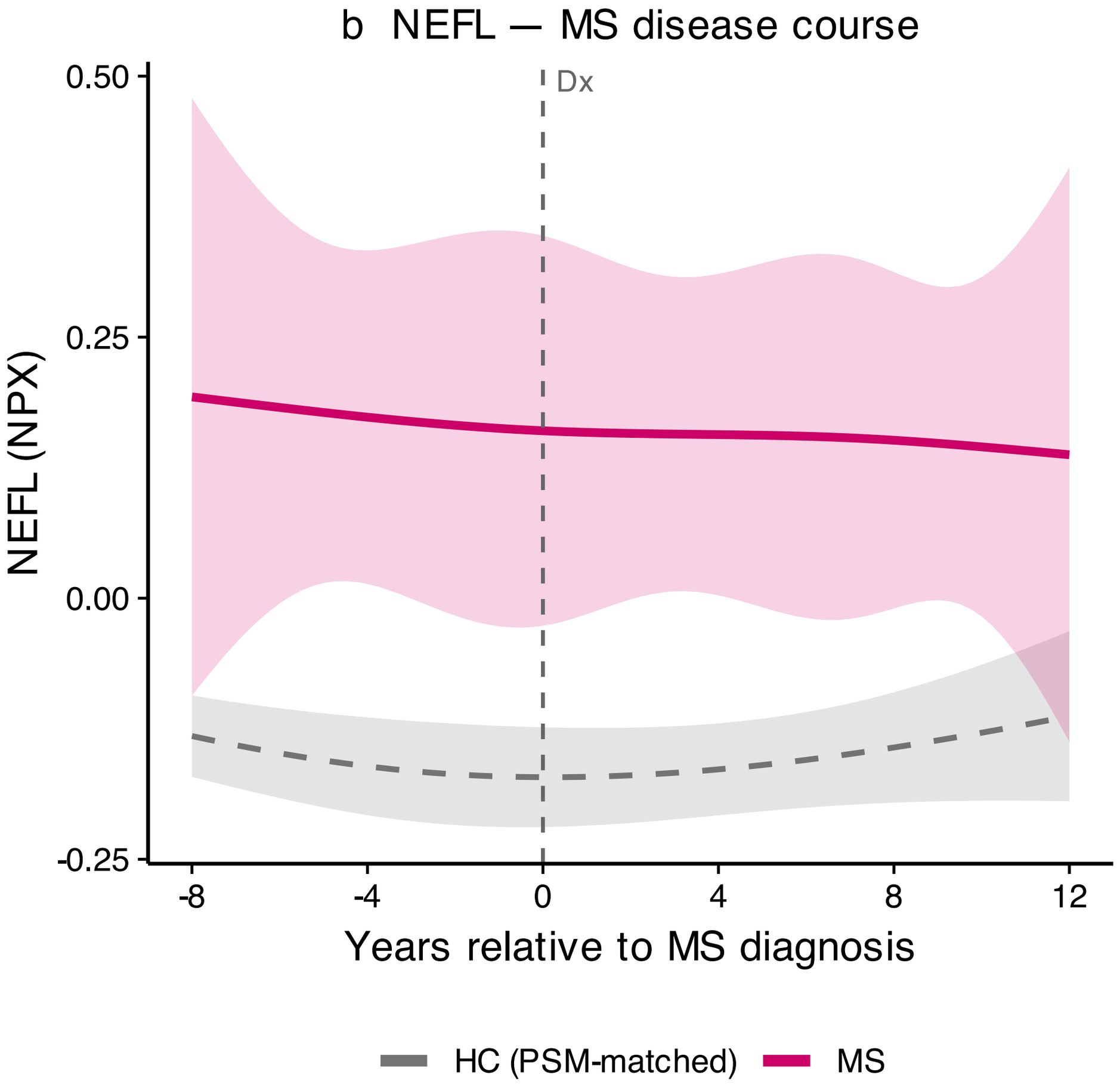
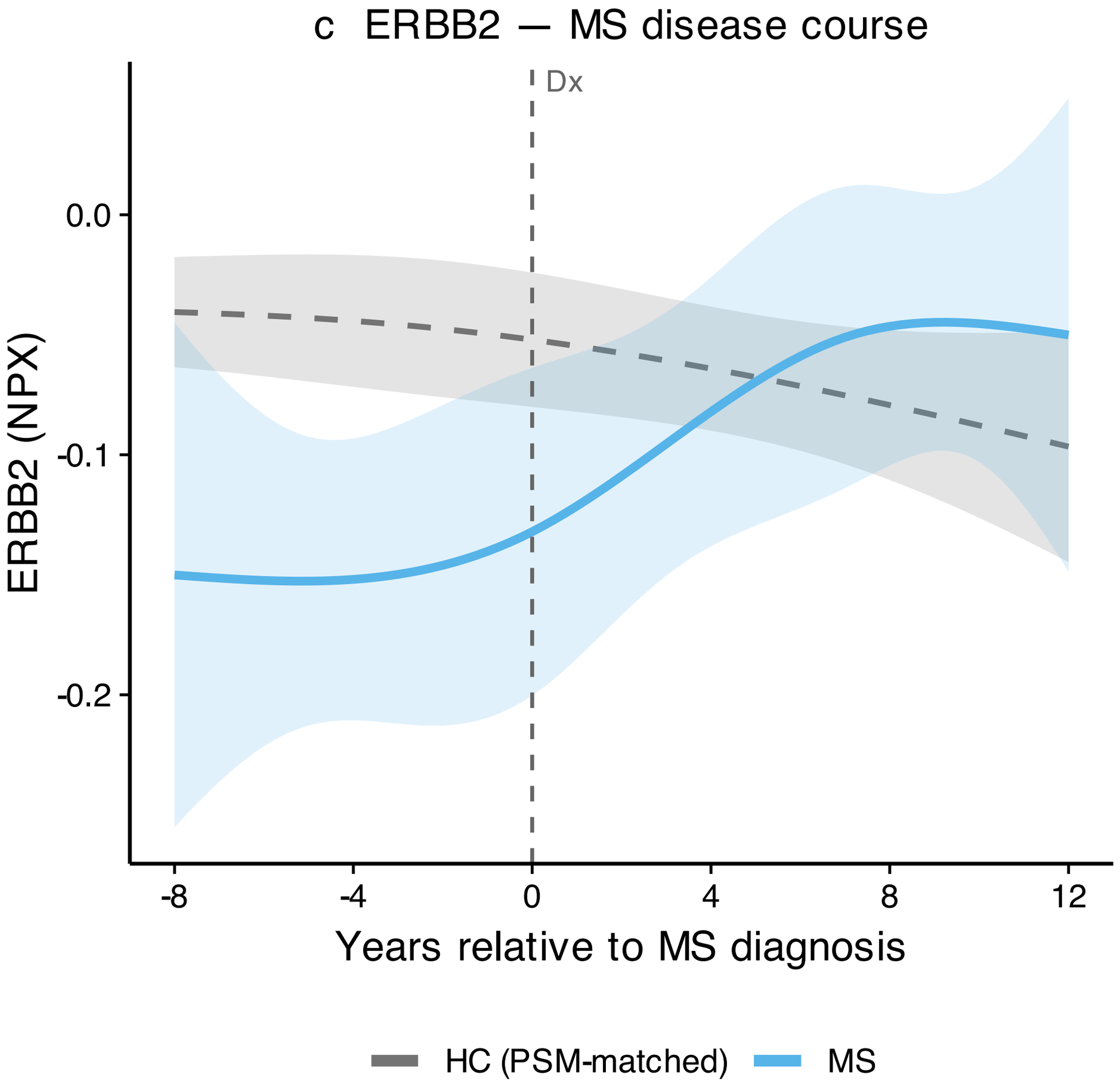
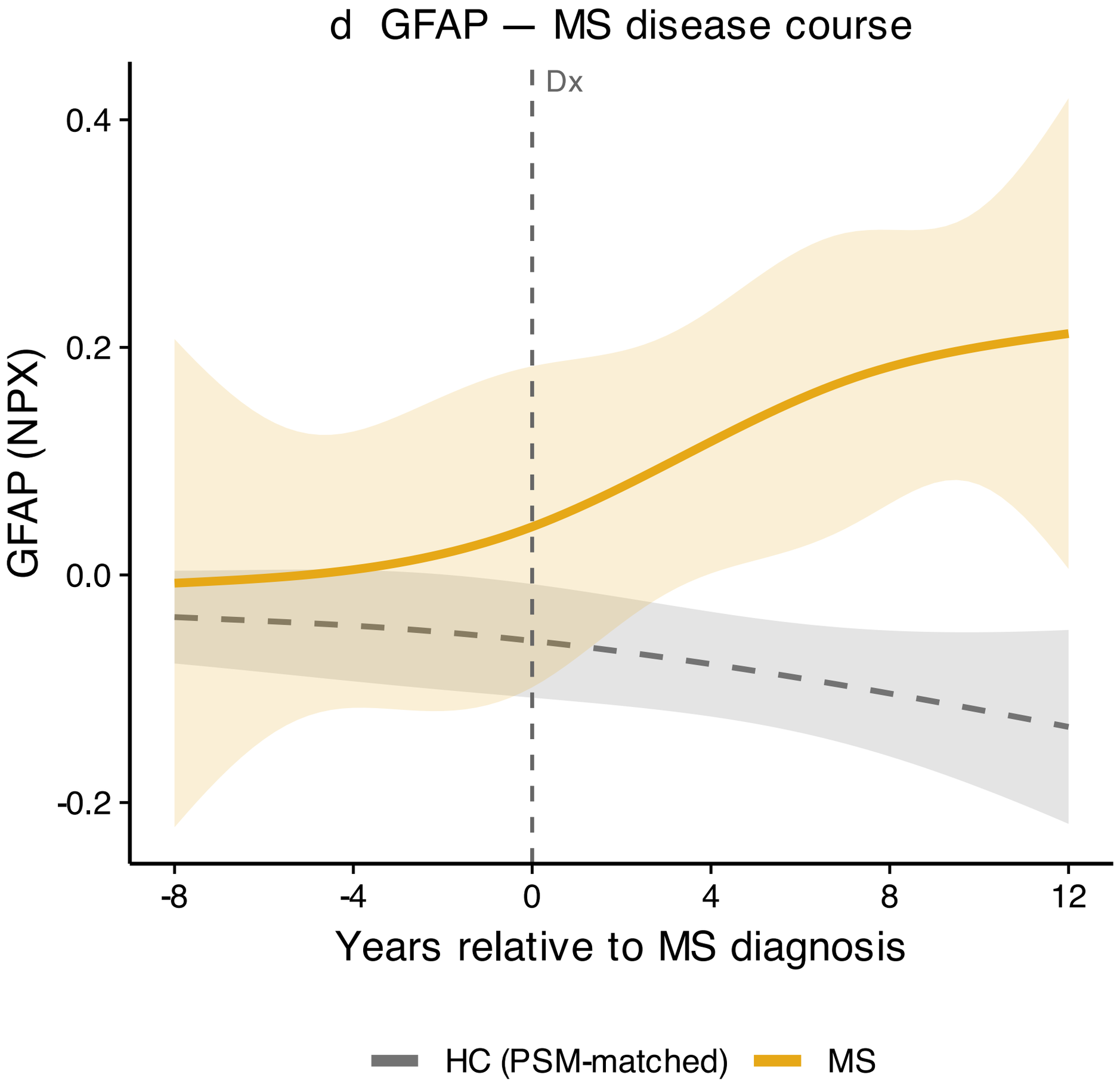
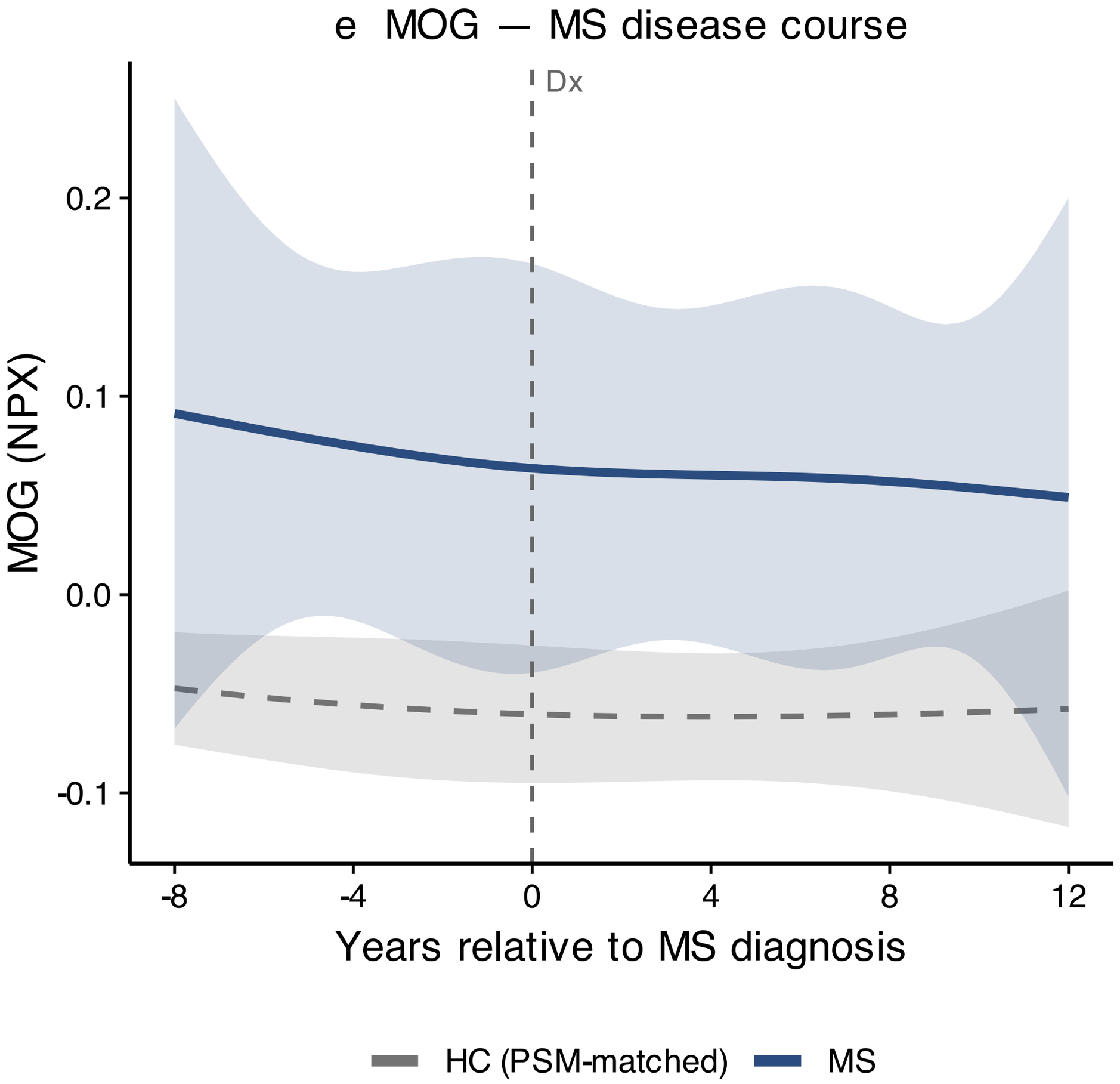
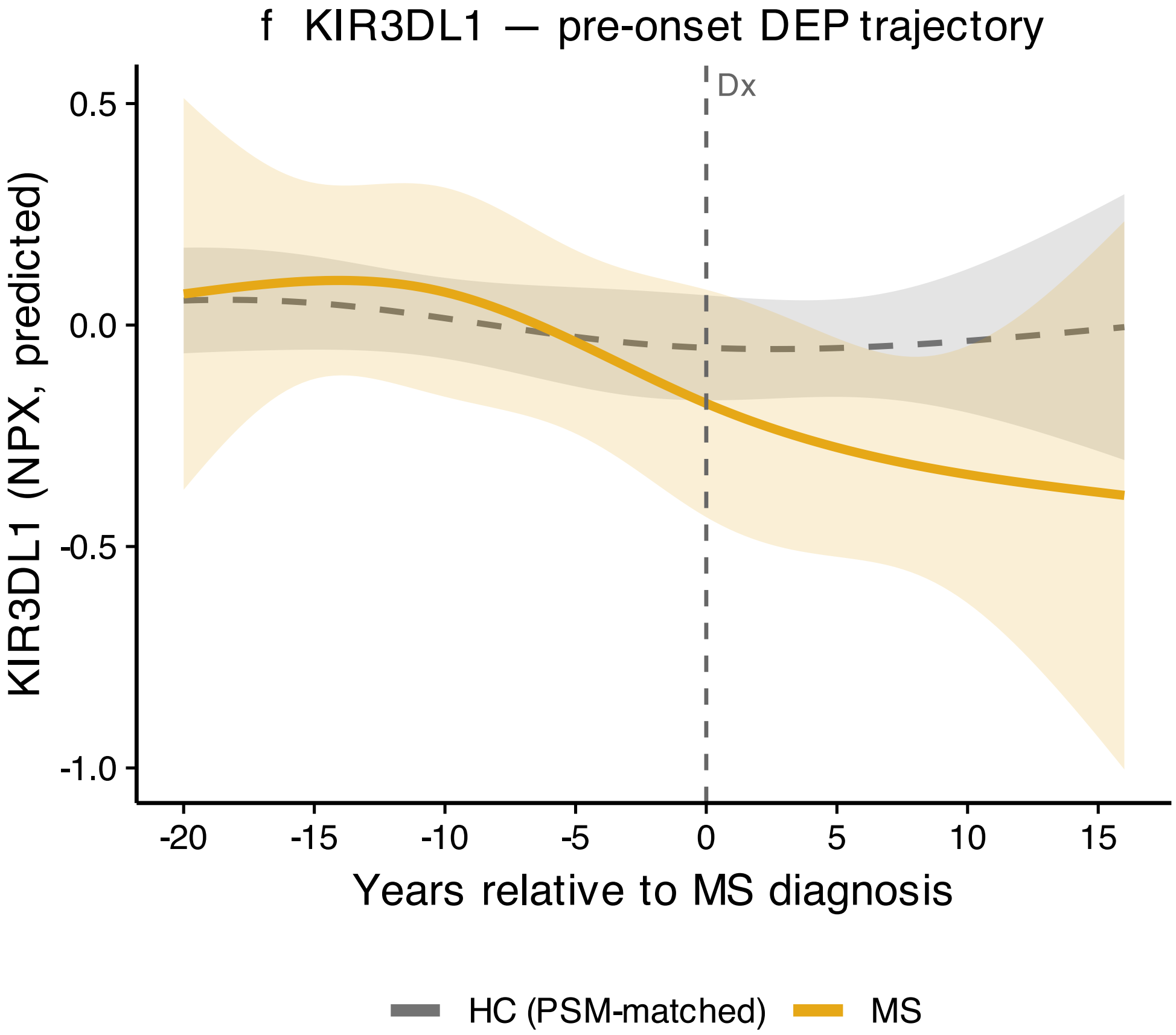

a MS sampling window
n=497 | 125 pre-onset, 372 post-onset
Dx
MS cases
0
5
10
15
20
-50
-40
-30
-20
-10
0
10
Years relative to MS diagnosis
Pre-onset
Post-onset

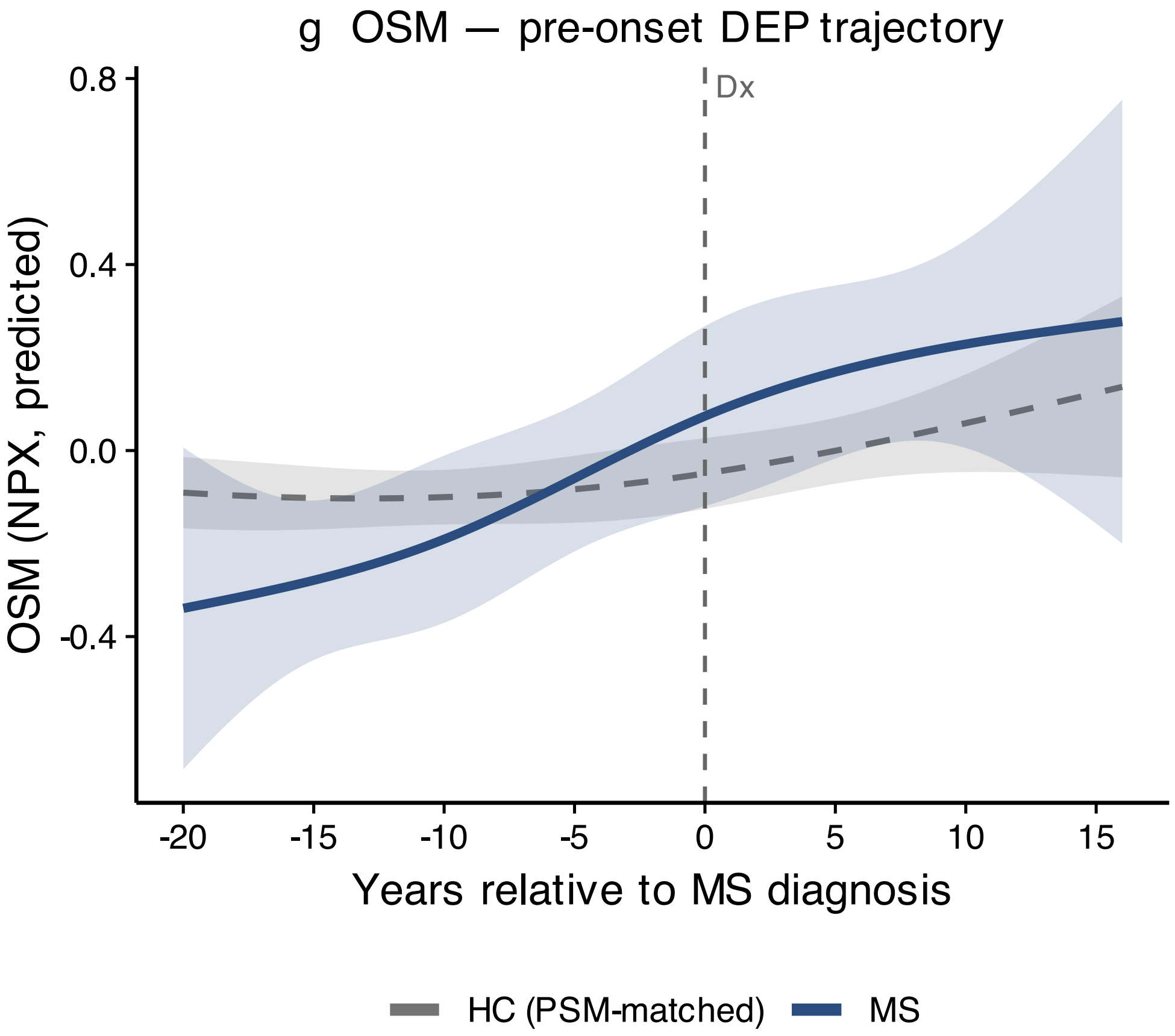

b NEFL — MS disease course
Dx
0.50
0.25
0.00
-0.25
NEFL (NPX)
-8
-4
0
4
8
12
Years relative to MS diagnosis
HC (PSM-matched)
MS

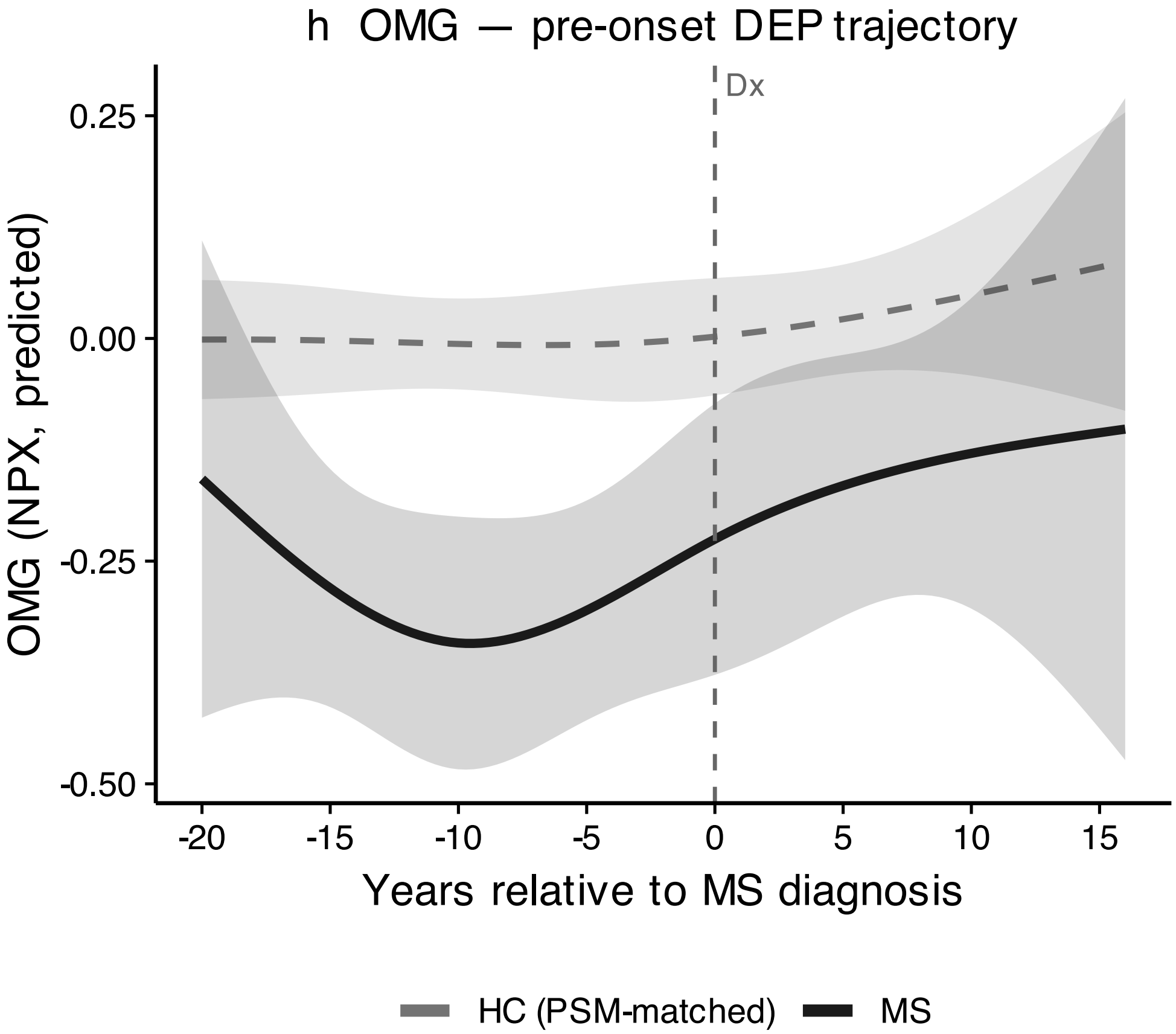

c ERBB2 — MS disease course
Dx
0.0
-0.1
-0.2
ERBB2 (NPX)
-8
-4
0
4
8
12
Years relative to MS diagnosis
HC (PSM-matched)
MS

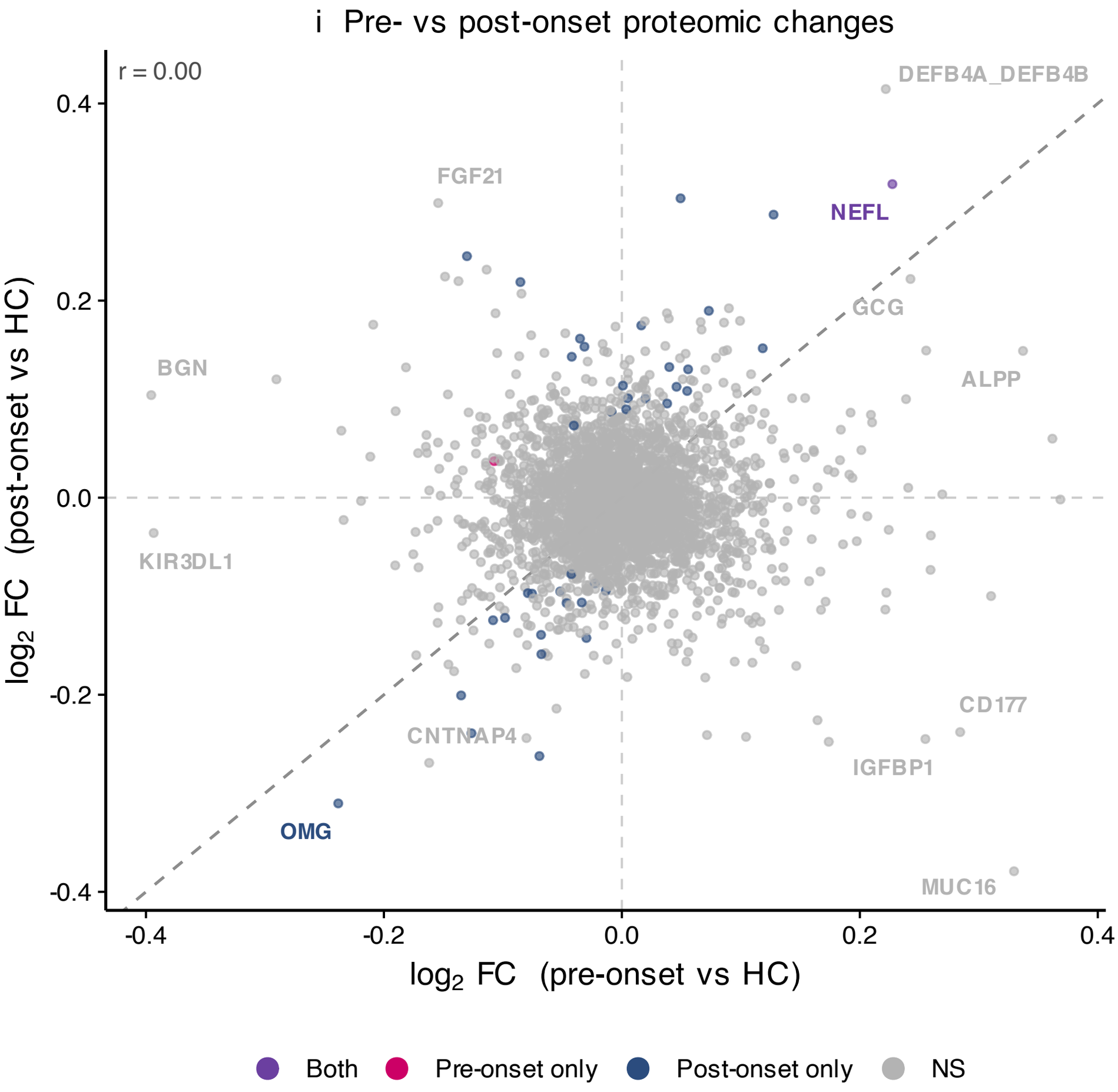

d GFAP — MS disease course
Dx
0.4
0.2
0.0
-0.2
GFAP (NPX)
-8
-4
0
4
8
12
Years relative to MS diagnosis
HC (PSM-matched)
MS

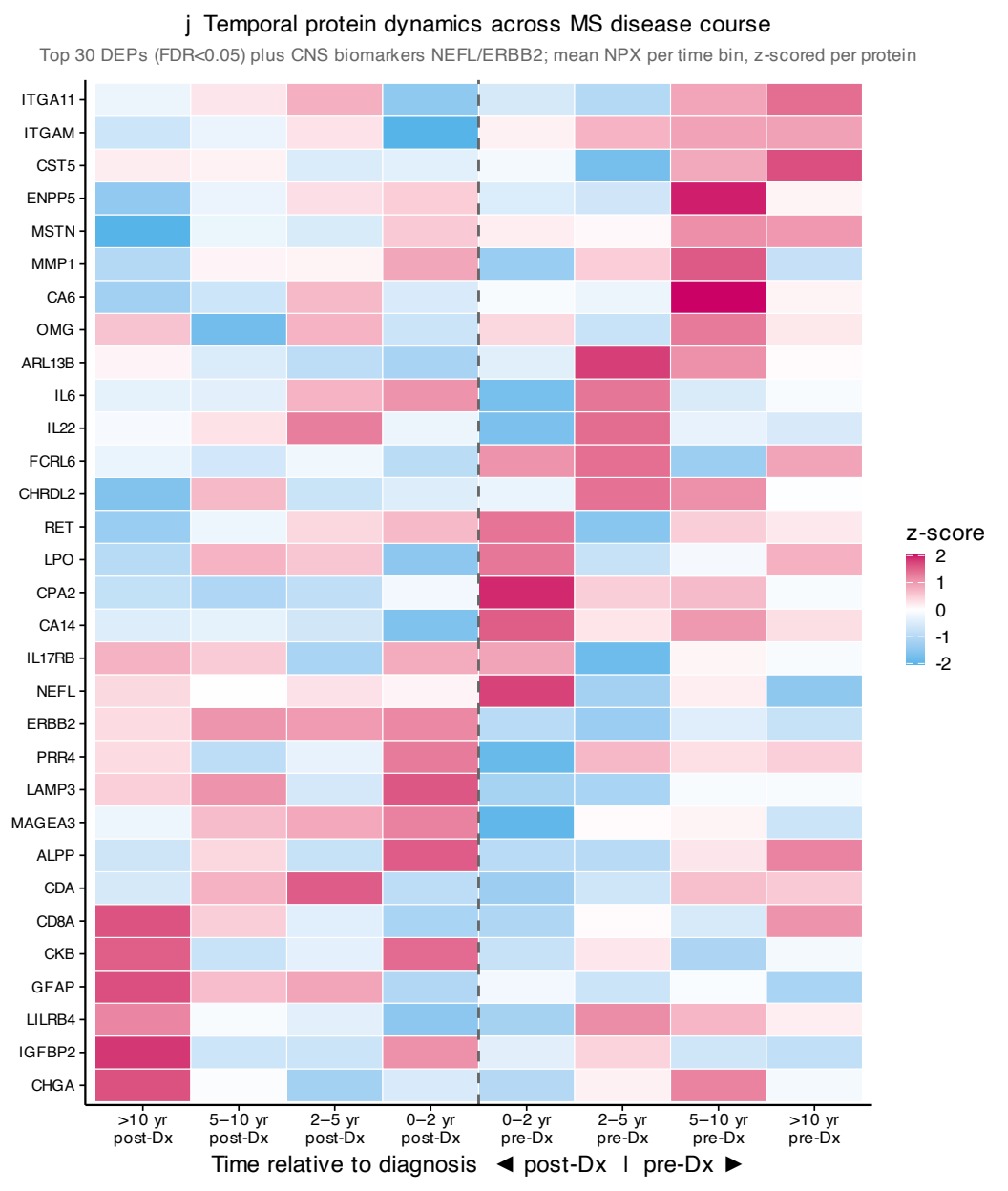

e MOG — MS disease course
Dx
0.2
0.1
0.0
-0.1
MOG (NPX)
-8
-4
0
4
8
12
Years relative to MS diagnosis
HC (PSM-matched)
MS

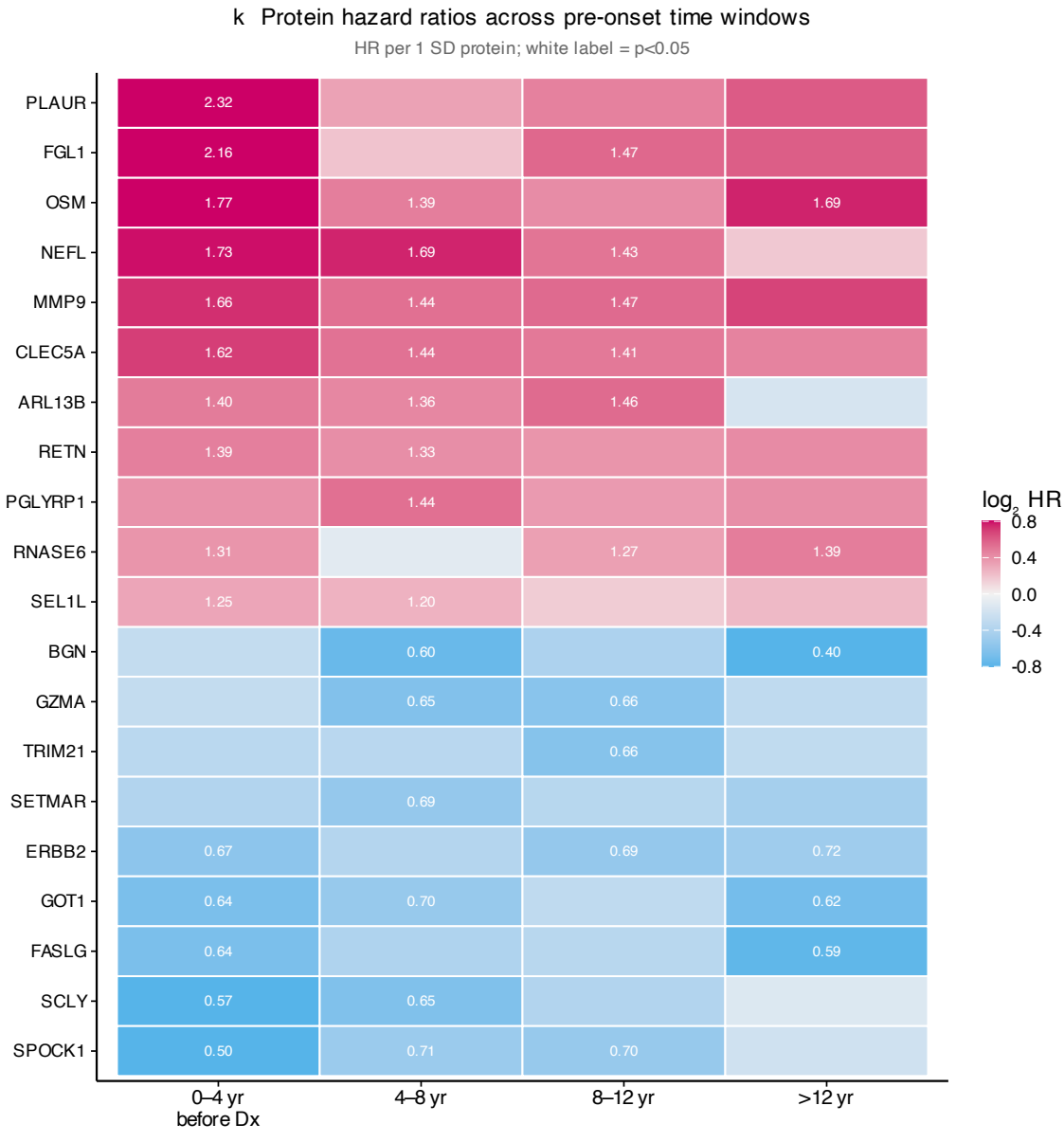

f KIR3DL1 — pre-onset DEP trajectory
Dx
0.5
0.0
-0.5
-1.0
KIR3DL1 (NPX, predicted)
-20
-15
-10
-5
0
5
10
15
Years relative to MS diagnosis
HC (PSM-matched)
MS


g OSM — pre-onset DEP trajectory
Dx
0.8
0.4
0.0
-0.4
OSM (NPX, predicted)
-20
-15
-10
-5
0
5
10
15
Years relative to MS diagnosis
HC (PSM-matched)
MS

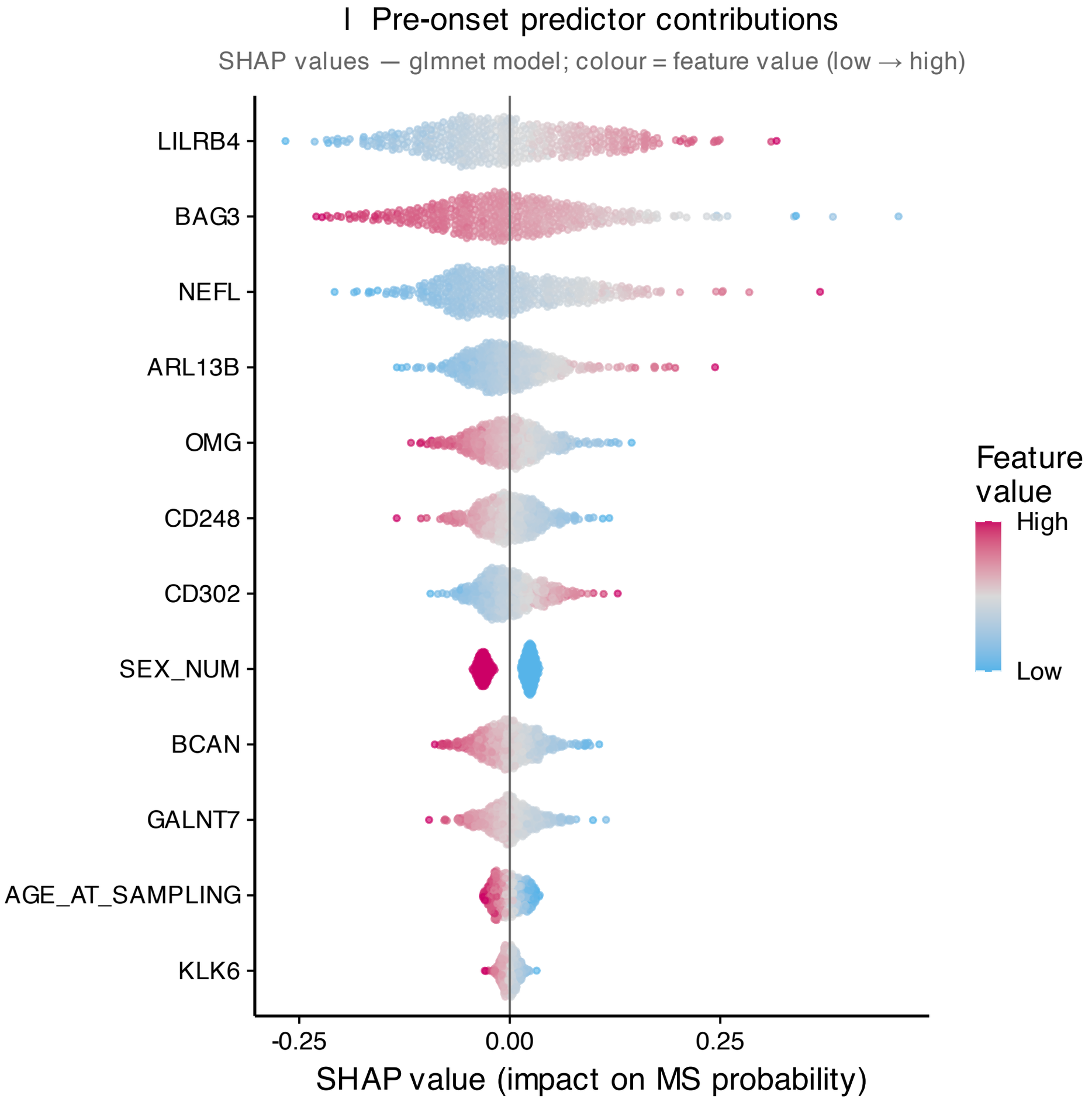

h OMG — pre-onset DEP trajectory
Dx
0.25
0.00
-0.25
-0.50
OMG (NPX, predicted)
-20
-15
-10
-5
0
5
10
15
Years relative to MS diagnosis
HC (PSM-matched)
MS

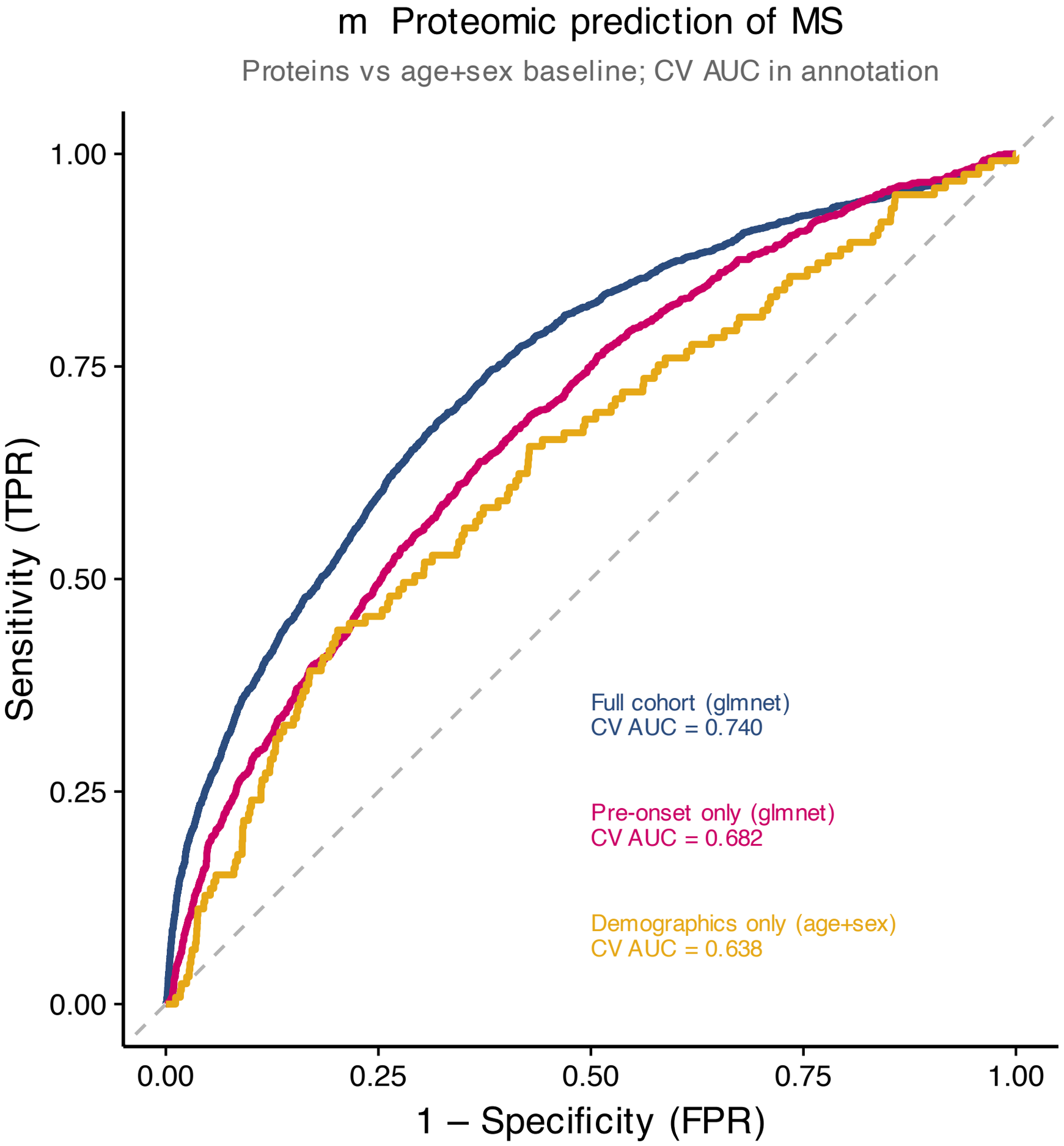

i Pre- vs post-onset proteomic changes
r = 0.00
DEFB4A_DEFB4B
FGF21
NEFL
GCG
BGN
ALPP
KIR3DL1
CD177
CNTNAP4
IGFBP1
OMG
MUC16
0.4
0.2
0.0
-0.2
-0.4
log2 FC (post-onset vs HC)
log2 FC (pre-onset vs HC)
Both
Pre-onset only
Post-onset only
NS

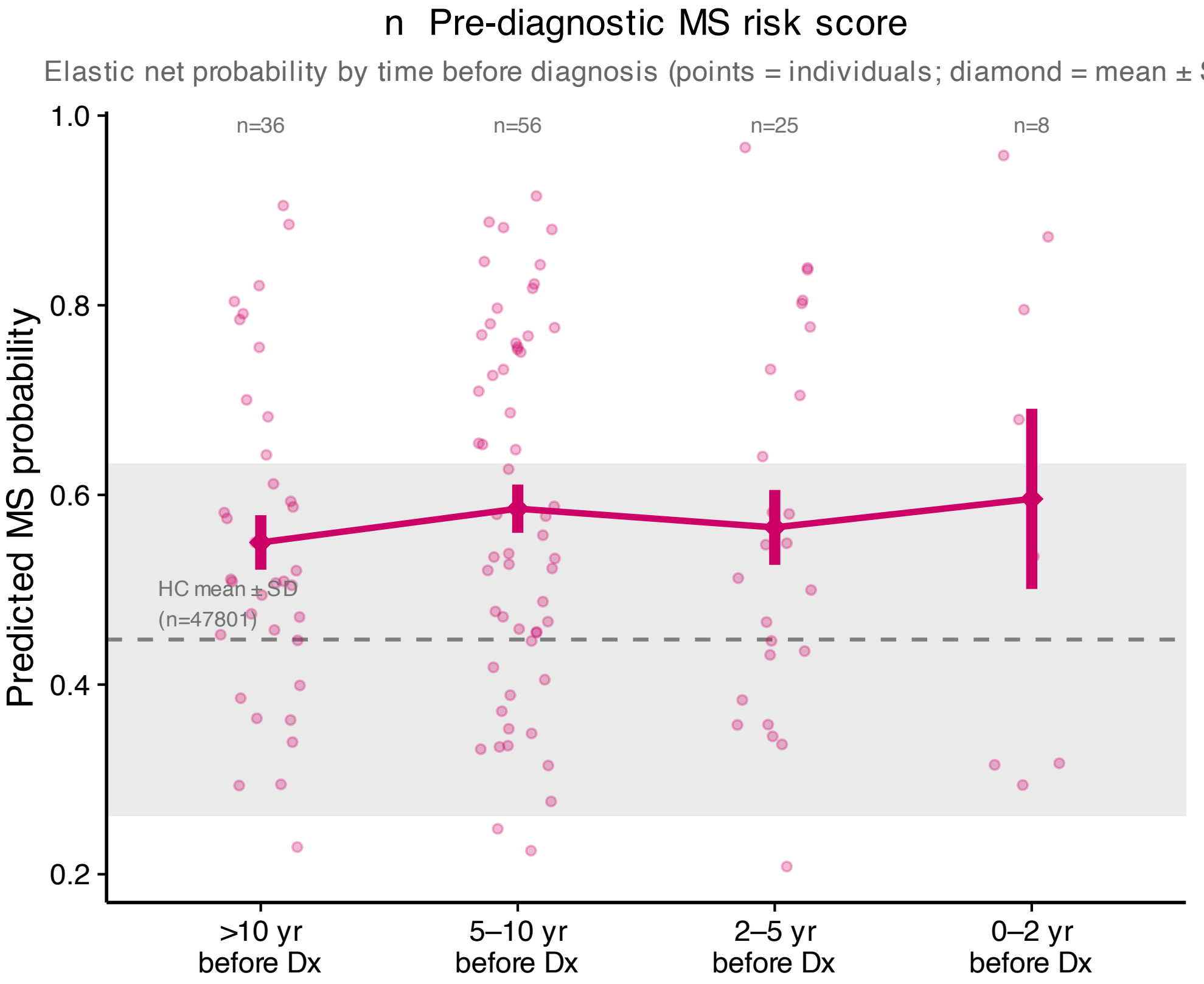

j Temporal protein dynamics across MS disease course
Top 30 DEPs (FDR<0.05) plus CNS biomarkers NEFL/ERBB2; mean NPX per time bin, z-scored per protein
ITGA11
ITGAM
CST5
ENPP5
MSTN
MMP1
CA6
OMG
ARL13B
IL6
IL22
FCRL6
CHRDL2
RET
LPO
CPA2
CA14
IL17RB
NEFL
ERBB2
PRR4
LAMP3
MAGEA3
ALPP
CDA
CD8A
CKB
GFAP
LILRB4
IGFBP2
CHGA
>10 yr post-Dx
5–10 yr post-Dx
2–5 yr post-Dx
0–2 yr post-Dx
0–2 yr pre-Dx
2–5 yr pre-Dx
5–10 yr pre-Dx
>10 yr pre-Dx
Time relative to diagnosis ◄ post-Dx | pre-Dx ►
z-score
2
1
0
-1
-2

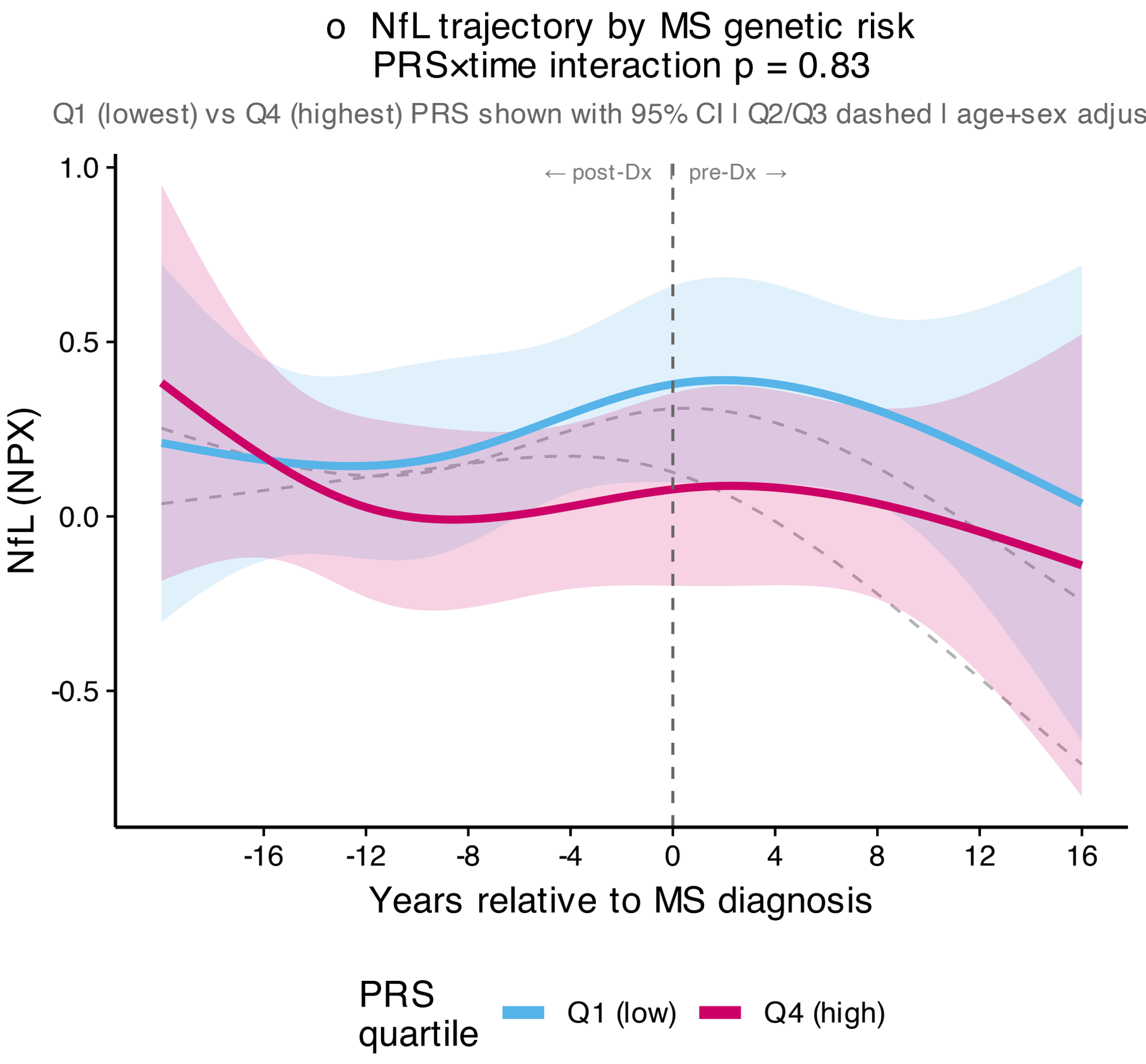


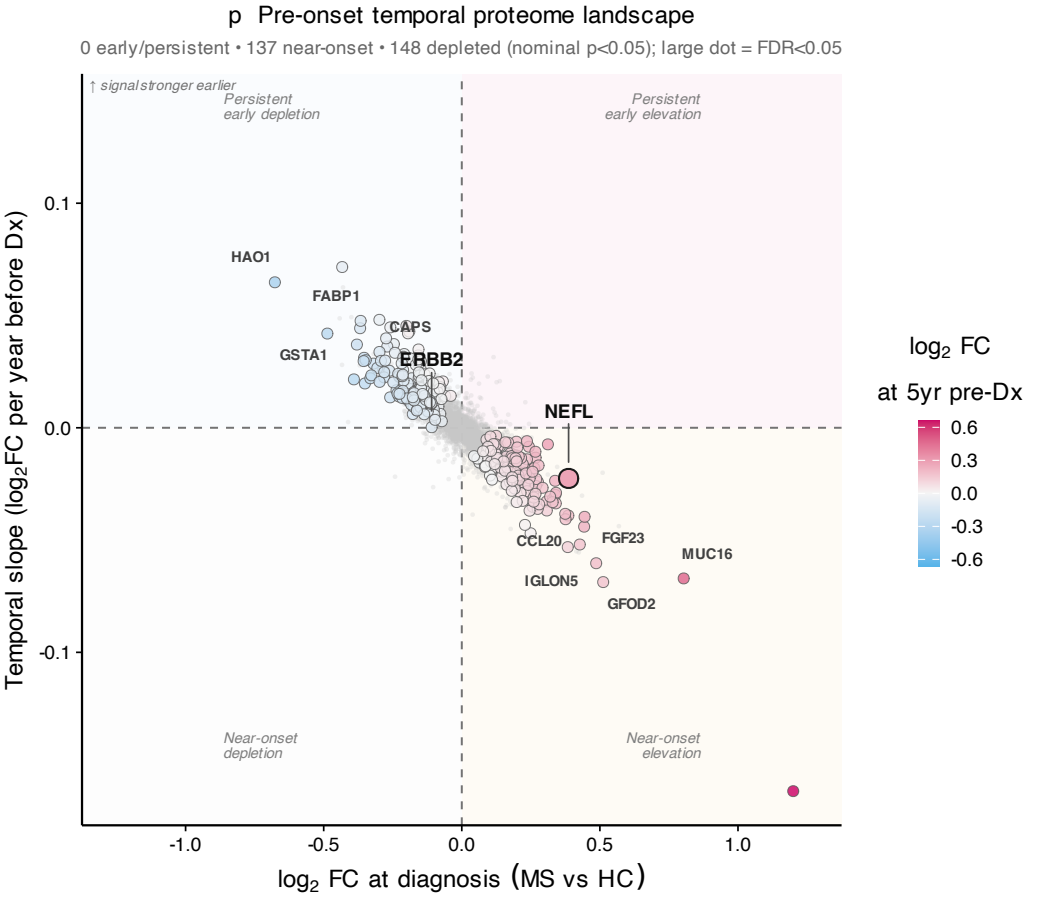
Cox model: all controls as censored events (standard for prospective cohorts) | age, sex, UMAP1/2 adjusted


I Pre-onset predictor contributions
SHAP values — glmnet model; colour = feature value (low → high)
LILRB4
BAG3
NEFL
ARL13B
OMG
CD248
CD302
SEX_NUM
BCAN
GALNT7
AGE_AT_SAMPLING
KLK6
-0.25
0.00
0.25
SHAP value (impact on MS probability)
Feature value
High
Low


m Proteomic prediction of MS
Proteins vs age+sex baseline; CV AUC in annotation
Sensitivity (TPR)
1 – Specificity (FPR)
0.00
0.25
0.50
0.75
1.00
Full cohort (glmnet)
CV AUC = 0.740
Pre-onset only (glmnet)
CV AUC = 0.682
Demographics only (age+sex)
CV AUC = 0.638


n Pre-diagnostic MS risk score
Elastic net probability by time before diagnosis (points = individuals; diamond = mean ±
Predicted MS probability
1.0
0.8
0.6
0.4
0.2
n=36
n=56
n=25
n=8
HC mean ± SD
(n=47801)
>10 yr
before Dx
5–10 yr
before Dx
2–5 yr
before Dx
0–2 yr
before Dx


o NfL trajectory by MS genetic risk
PRS×time interaction p = 0.83
Q1 (lowest) vs Q4 (highest) PRS shown with 95% CI | Q2/Q3 dashed | age+sex adjus
← post-Dx
pre-Dx →
1.0
0.5
0.0
-0.5
NfL (NPX)
-16
-12
-8
-4
0
4
8
12
16
Years relative to MS diagnosis
PRS quartile
Q1 (low)
Q4 (high)


p Pre-onset temporal proteome landscape
0 early/persistent • 137 near-onset • 148 depleted (nominal p<0.05); large dot = FDR<0.05
↑ signal stronger earlier
Persistent early depletion
Persistent early elevation
Near-onset depletion
Near-onset elevation
HAO1
FABP1
CAPS
GSTA1
ERBB2
NEFL
CCL20
FGF23
MUC16
IGLON5
GFOD2
Temporal slope (log2FC per year before Dx)
log2 FC at diagnosis (MS vs HC)
0.1
0.0
-0.1
-1.0
-0.5
0.0
0.5
1.0
log2 FC
at 5yr pre-Dx
0.6
0.3
0.0
-0.3
-0.6